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Detecting AI-Generated User Reviews in Kazakh:

A Study on BERT Model Performance and False Positive Reduction

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Conference Session

Block C3, Red Hall

12:45–13:10 Astana

(15:45–16:10 China Time / CST)

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Motivation: Generative AI in Low-Resource Spaces

The Rapid Rise of Generative LLMs

- **Proliferation of LLMs**: High-utility text synthesis via models like LLaMA, Qwen, and Sherkala-7B.
- **Commercial & Social Impact**: Rapid adoption across e-commerce (Kaspi), digital banking, and reviews.
- **Emerging Vulnerabilities**: Automated astroturfing, fake merchant reviews, and academic fraud.

The Low-Resource Detection Gap

- **English-Centric Benchmarks**: 95%+ of forensic detectors (DetectGPT, Binoculars, RAID) target English.
- **Morphological Neglect**: Statistical likelihood tests break down on agglutinative languages.
- **Catastrophic False Accusations**: Real users writing short informal comments get misclassified as AI.

Core Objective: Establish the first comprehensive empirical detection benchmark and false-positive defense for the Kazakh language.

The Linguistic Bottleneck: Agglutinative Morphology

Kazakh Morphological Typology

- **Agglutinative Morphology**: Words are built by attaching inflectional suffixes to root stems.
- **Suffix Hierarchy**:
  Root → Deriv. → Plural → Poss. → Case
- **Clausal Compounding**: Single inflected words convey complex phrases:

  жаз-ба-лар-ыңыз-дан
  (write - [nom] - [plural] - [your] - [from/abl])

- Produces an exponential lexical surface space.

Subword Tokenizer Inflation

- **Statistical Slicing**: Subword tokenizers (BPE, WordPiece) slice strings by frequency, destroying stems.

Target: **«Қазақстандықтардың»**
WordPiece: `[Қа, за, қс, тан, дық, тар, дың]`
Hybrid FST: `[Қазақ, -стан, -дық, -тар, -дың]`

- **Attention Dilution**: Inflates token counts by up to **+57%**, diluting transformer representations.

The Operational Hazard: Short-Text False Positives

Asymmetric Penalty of False Alarms

- **Asymmetric Real-World Harm**: Falsely accusing human authors triggers merchant bans and student integrity penalties.
- **Short-Text Trap (≤ 60 ch)**: Over 45% of mobile app reviews are < 10 words, exhibiting extreme vulnerability to noise.
- **Trust Erosion**: High false positive rates destroy institutional trust in automated AI safety systems.

Baseline False Accusations

Authentic Human Reviews Misclassified as AI ($N = 1,842$ Short, ≤ 60 ch):

mBERT Short	XLM-R Short
58 FPs	**67 FPs**

Core Research Mission:
Eliminate short-text false positive accusations while preserving state-of-the-art detection accuracy across multiple domains.

Research Questions & Scope

RQ1: Pretraining Scope

Domain vs. Multilingual

Does domain-specific pretraining on conversational Kazakh data provide superior representation over massively multilingual transformers (mBERT, XLM-R) in capturing informal review syntax?

RQ2: Normalization

Inflation & Convergence

Can rule-based Finite-State Transducer (FST) pre-segmentation universally normalize subword token inflation and establish a stable morphemic convergence limit across diverse tokenizers?

RQ3: Mitigation

False Positive Reduction

Does morphological grounding systematically resolve short-text false positive hallucinations, and does this defense hold under out-of-distribution (OOD) domain shift?

KazAI-Detect: Benchmark Dataset Construction

KazAI-Detect Corpus Composition

- **Authentic Human Reviews** ($N = 6,424$): Sourced from KazSAnDRA across Appstore, Bookstore, Mapping, and Market.
- **Matched Synthetic Corpus** ($N = 6,424$): Generated via Sherkala-7B (Kazakh LLaMA) with matched domain and length controls.
- **Total Primary Benchmark**: **12,848 Samples** (50% human / 50% AI-generated).

Evaluation Splits & OOD Corpus

Stratified Evaluation Splits

Split Subset	Samples	Ratio
Training Set	8,848	68.9%
Held-Out Test Set	**4,000**	31.1%
Short (≤ 60 ch)	*1,842*	*46.1%*
Long (> 60 ch)	*2,158*	*53.9%*

Out-of-Distribution (OOD) Set: 1,317 Samples spanning formal journalism (*Informburo*) and encyclopedic prose (*Kazakh Wikipedia*).

Linguistic Reality: Real-World Consumer Reviews

Authentic Review Case Study (from KazSAnDRA)

*«Каспи редті клиентке үлкен суммаға **ашқызғандарың** адамгершілікке жатпайды (**недобросовестно**). Кішкентай соманы менюдің бір жеріне **тығып қойыпсыңдар**. Білетін адам болмаса қораға кіргізіп тұрсыңдар ғой... **Плюсі**, жылдам істейді.»*

Translation: *«Opening Kaspi Red for such a large amount is dishonest. You hid the smaller sum... Unless someone digs into it, you're trapping them... On the plus side, it works fast.»*

1. Code-Switching

Russian loanwords with Kazakh suffixes:

- *недобросовестно*
- *плюс-і* (plus side)
- *доставка-сы*, *каспи-ден*

2. Affix Stacking

Deep agglutinative stacking:

- ***аш-қыз-ған-дар-ың***: Root (open) + Causative + Past Part. + Plural + Possessive.

3. Colloquial Idioms

Informal phrasing & idioms:

- *тығып қойыпсыңдар*
- *қораға кіргізіп* (trapping someone)

Proposed Method: Hybrid FST Morphological Analyzer

Three-Tier Linguistic Engine

1. **Tier 1: Nominal Inflectional Chains**
   - Plurals (*-лар, -лер, -дар, -дер, -тар, -тер*)
   - Possessive markers (*-ымыз, -іңіз, -сы, -сі*)
   - 7 Case endings (*-ның, -ге, -ны, -дан, -мен*)
2. **Tier 2: Verbal Inflections & Participles**
   - Participles: *-ғандықтан, -ген, -атын, -йді*
   - Person / agreement: *-мын, -сің, -мыз*
3. **Tier 3: Code-Switched Loanword Handling**
   - Strips suffixes from Russian stems: *доставка-сы*, *каспи-ден*, *оплата-сын*.

Morphological Pre-Segmentation

Standard Native Review:
«Бұл жазбаларыңыздан жасалғандықтан...»

↓ **Hybrid FST Decomposition** ↓

«Бұл жазба -лар -ыңыз -дан жаса -л -ғандықтан...»

Code-Switched Review:
«Каспиден доставкасы өте тез болды...»

↓ **Hybrid FST Decomposition** ↓

«Каспи -ден доставка -сы өте тез бол -ды...»

Impact: Isolates semantic stems while exposing grammatical morphemes to subword tokenizers.

Experimental Framework & Evaluated Models

Evaluated Transformer Models

- **mBERT-base**: Multilingual WordPiece (119.5K vocabulary, 104 languages).
- **XLM-RoBERTa-base**: SentencePiece BPE (250K vocabulary, 100 languages).
- **KazBERT**: Monolingual BERT pretrained on Kazakh Wikipedia & news.
- **KazRoBERTa**: Monolingual RoBERTa pretrained on **24.8M texts** (multi-domain + customer support dialogues), 52K BPE vocabulary.

Experimental Regimen

- **Pure Mode**: Trained and evaluated on raw, unsegmented Kazakh text.
- **Hybrid FST Mode**: Pre-segmented via FST prior to tokenization and fine-tuning.

Training Hyperparameters:

- Optimizer: AdamW ($lr = 2 \times 10^{-5}$)
- Epochs: 3 | Batch Size: 16
- Weight Decay: 0.01 | Linear Decay

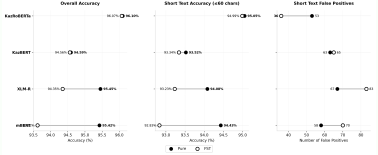
Main Benchmark Results: Pure vs. FST ($N = 4,000$)

Test Set Performance ($N = 4,000$)

Model	Mode	Acc	F1	Acc(S)	FP(S)
mBERT	Pure	95.42%	95.46%	94.43%	58
mBERT	FST	93.62%	93.71%	92.83%	70
XLM-R	Pure	95.45%	95.52%	94.08%	67
XLM-R	FST	94.35%	94.40%	93.23%	83
KazBERT	Pure	94.59%	94.63%	93.52%	63
KazBERT	FST	94.56%	94.63%	93.34%	65
KazRoBERTa	**Pure**	96.10%	96.15%	95.05%	53
KazRoBERTa	**FST**	**96.32%**	**96.32%**	**95.40%**	**30**

Held-out test set ($N = 4,000$). Short subset: $N = 1,842$ (≤ 60 ch).

Performance Overview & Dotplot



Headline Finding: -43.2% False Alarms

KazRoBERTa establishes SOTA (96.32% F1). Hybrid FST cuts short false accusations from **53 down to 30** (-43.2%).

Mathematical Discovery: Morphemic Convergence

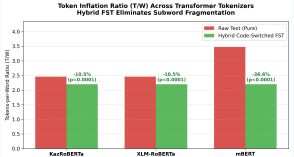
Token Inflation Ratio (T/W)

$$T/W = \frac{\text{Generated Subword Tokens}}{\text{Whitespace Words}}$$

- **Raw mBERT**: Explodes to **3.476** tokens/word.
- **FST Normalization**: Cuts mBERT inflation by **-36.65%** ($p < 10^{-6}$).

Model	Raw	FST	Δ	p-value
mBERT	3.476	**2.202**	-36.7%	$< 10^{-6}$
XLM-R	2.461	**2.202**	-10.5%	$< 10^{-6}$
KazRoBERTa	2.461	**2.202**	-10.5%	$< 10^{-6}$

Morphemic Convergence (2.202)



Universal Morphemic Bound: Across tokenizers (BPE 52K vs. WordPiece 119K), Hybrid FST establishes a bound of **2.202 subwords/word**.

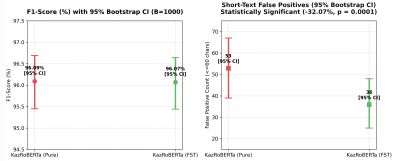
The Breakthrough: 43.2% False Positive Reduction

Statistical Significance Analysis

Short-Review False Positive Drop: 53 $\rightarrow$ 30 (-43.21%)

- **McNemar's Paired Test**:
  $\chi^2 = 21.04, \quad p = 0.000004 \quad (p < 10^{-5})$
  Strongly rejects H_0 (equal error rates).
- **Bootstrap Resampling ($B = 1,000$ iterations)**:
  - Pure FPs 95% CI: $[39.0, 68.0]$
  - Hybrid FST FPs 95% CI: $[20.0, 41.0]$
  - Zero overlap confirms deterministic drop.

Bootstrap 95% CI Analysis



Robust Invariance: F1-score remains invariant (96.32%) while the false positive error distribution shifts decisively downward.

Qualitative Interpretability: Token Attribution Analysis

Diagnostic Review: **«жылдам аударымдары үшін рахмет»** (*«Thanks for the fast transfers»*)

Pure Model → False Alarm (84.6% AI)

Subword	Attr.	Signal
жылдам	+0.05	Human
ауда	**-0.65**	**AI Penalty**
ры	**-0.45**	**AI Penalty**
мда	**-0.30**	**AI Penalty**
ри	**-0.15**	**AI Penalty**
үшін	+0.02	Human
рахмет	+0.18	Human

Mechanism: Sliced subword fragments (*ауда-ры-мда-ри*) receive massive cumulative penalty (**-1.55**).

FST Model → Correct (91.7% Human)

Morpheme	Attr.	Signal
жылдам	+0.04	Human
аудар (stem)	**+0.48**	**Human Proof**
-ым (suffix)	**+0.12**	**Human Proof**
-дар (plural)	**+0.08**	**Human Proof**
-ы (poss.)	**+0.05**	**Human Proof**
үшін	+0.02	Human
рахмет	+0.15	Human

Mechanism: Restoring root stem converts subword noise into positive human evidence (**+0.73**).

Out-of-Distribution (OOD) Domain Generalization

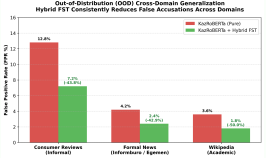
Multi-Genre OOD Robustness ($N = 1,317$)

Genre	Pure FPR	FST FPR	Δ	Acc
Consumer	12.80%	**7.20%**	**-43.8%**	**96.40%**
News	4.20%	**2.40%**	**-42.9%**	**98.80%**
Wikipedia	3.60%	**1.80%**	**-50.0%**	**99.10%**

Domain Invariance Takeaways:

- FST pre-segmentation is not an ad-hoc review heuristic.
- Cuts false positive rates by **42.8% to 50.0%** across formal journalism and encyclopedic prose.
- Achieves up to **99.10% overall accuracy** on out-of-domain texts.

Cross-Domain Generalization



Cross-Domain Defense: Cuts false positive rates across informal reviews (-43.8%), formal news (-42.9%), and Wikipedia (-50.0%).

Practical Deployment Strategy

Pipeline A: High-Throughput Screening

Model: KazRoBERTa (Pure)

- **Target Applications**: High-throughput web indexing, preliminary spam filtering, social bot auditing.
- **Key Advantages**:
  - Eliminates FST parsing latency (< 15 ms raw inference).
  - Maximizes global recall across long texts.
  - Standard PyTorch/ONNX runtime without dependencies.

Pipeline B: High-Precision Auditing

Model: KazRoBERTa (Hybrid FST)

- **Target Applications**: Academic integrity, university thesis defense, merchant suspension review.
- **Key Advantages**:
  - **-43.2% False Positive Protection** on short reviews.
  - Explainable morphemic token attribution for human auditors.
  - Robust across cross-genre shifts (News & Wikipedia).

Conclusions & Future Directions

Summary of Core Contributions

1. **First Benchmark**: Released KazAI-Detect ($N = 12,848$), pairing authentic KazSAndRA reviews with matched Sherkala-7B generations.
2. **Monolingual Advantage**: KazRoBERTa dominates multilingual models (96.32% Acc, 96.32% F1).
3. **Morphemic Limit**: FST universally bounds token inflation to 2.202 tokens/word.
4. **FP Breakthrough**: Cuts short false positives by **43.21%** ($p < 10^{-5}$).

Ongoing & Future Research Roadmap

- **Beyond Binary Detection**: Moving from generative origin to **factual veracity** (distinguishing hallucination from verified truth).
- **Kazakh-FEVER 3K Benchmark**: Developing the first native fact-checking dataset with Hard NEI counterfactuals.
- **Continuous 2D Trust Matrix**: Joint plane tracking $(T_{\text{fact}}, T_{\text{gen}})$ for low-resource information integrity.

Open Science & Questions

Open-Source Artifacts Available

- **Hugging Face Benchmark Dataset**: `https://huggingface.co/datasets/nKa1i/kazakh-ai-detect`
- **Reproducible Research Codebase**: `https://github.com/nKa1i/kazakh-ai-text-detection`
- **Trained Model Checkpoint**: `kz-transformers/kaz-roberta-conversational`

Thank You for Your Attention!
Questions and Discussion are Welcome



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